Lucent Biotech

B2R Medical Application – Business Requirements & Functional Design Document

LUCENT BIOTECH
LIMITED

1. Project Overview

Lucent Biotech, a leading pharmaceutical manufacturer specializing in third-party medicine production, currently operates primarily through a B2B (Business-to-Business) model. To strengthen its market footprint, improve operational efficiency, and establish a direct connection with retailers, the company is now planning to launch a B2R (Business-to-Retail) mobile application.

This platform will allow retailers including medical stores, distributors, and pharmacies to purchase medicines directly from Lucent Biotech. The system will be supported by Agents (Medical Representatives / MRs) who manage retailer relationships and generate sales through a commission-based referral model.

The application will function as a digital pharmaceutical marketplace, streamlining product discovery, order placement, and documentation while improving transparency, operational efficiency, and overall marketing reach.

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2. Stakeholders / User Roles

The ecosystem will involve three primary user roles:

a. Admin (Lucent Biotech Team)
b. Retailer (Medical Store / Distributor / Pharmacy)
c. Agent / MR (Medical Representative)

Each role will have distinct access rights and functionalities to support the overall B2R operation.

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3. Business Objectives

a. Transform Lucent Biotech's business from pure manufacturing to direct retail engagement through a digital B2R platform.
b. Streamline retailer and agent on boarding via digital verification and approval workflows.
c. Drive sales and product marketing through MRs and in-app promotions.
d. Enable order management, inventory control, and delivery tracking in one unified system.
e. Establish commission-based agent performance tracking.
f. Offer data-driven insights for management and operational optimization.
g. Strengthen Lucent Biotech's market visibility and customer engagement.

4. Business Model Summary

Model Type: Third Party Manufacturing & Direct-to-Retail Distribution (B2R)

a. Operational Workflow:
   i. Lucent Biotech manufactures medicines under its own and partner brands.
   ii. Retailers register, submit necessary documents (Drug License, GST, PAN), and once approved can order medicines directly through the app.
   iii. Agents (MRs) help onboard retailers, promote products, and earn commissions based on orders or successful referrals.
   iv. Admin (Lucent Team) manages the ecosystem products, pricing, users, orders, and commissions from a centralized dashboard.

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5. Application Architecture

Platform:

a. Mobile Application: Android + iOS
b. Admin Web Portal: Responsive dashboard for Lucent Biotech team.
c. Hosting: AWS Cloud Infrastructure.
d. EC2 (Application Server), RDS (Database), S3 (Storage for documents and images), CloudFront (CDN for speed and security), Email service, Push notifications.

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6. Functional Modules

a. Admin Portal (Lucent Biotech Backend)

i. Dashboard Overview
ii. Real-time insights and KPIs:
   1. Total Retailers, Active/Pending/Approved
   2. Total Agents (MRs)
   3. Total Orders, Sales Revenue, Product Performance
   4. Pending Approvals & Payouts
   5. Commission Statistics
   6. Low-stock alerts
   7. Notifications & Reminders

iii. User Management

1. Retailer Management

a. View all registrations with status (Pending/Approved/Rejected)
b. Review submitted documents (Drug License, GST, PAN)

c. Approve or reject applications
d. Deactivate/reactivate retailers
e. Region-based grouping for better control

2. **Agent (MR) Management**
    a. Manage MR profiles and documents (ID proof, bank details)
    b. Approve or reject MRs post verification
    c. Assign MRs to specific territories or product categories
    d. Monitor performance metrics

iv. **Access Roles in Admin Panel**
1. Super Admin: Full access
2. Operations Manager: Orders, Stock, Product management
3. Finance Admin: Commission, Payment reconciliation
4. Support Executive: Retailer queries and tickets
5. Marketing Admin: Campaigns, Offers, and Communication tools

v. **Product Management**
1. Add/Edit/Delete product listings with details:
    a. Name, Composition, MRP, Pack Size, Manufacturer, Expiry, Product Images
2. Product categorization (Tablets, Syrups, Capsules, etc.)
3. Real-time stock management
4. Region-based visibility settings
5. Bulk product upload via CSV/Excel
6. Control product activation/deactivation

vi. **Order Management**
1. Centralized order view (Pending → Processing → Shipped → Delivered)
2. Assign warehouse/distributor for dispatch
3. Auto-generate invoices
4. Manage returns, replacements, and refunds
5. Downloadable order reports

vii. **Commission Management**
1. Set commission rates per:
    a. Agent (MR)
    b. Product/Category
    c. Retailer onboarding or order milestone
2. Automated calculation and commission ledger
3. Approve/Reject/Release commissions
4. Track referral performance
5. Generate downloadable commission reports

viii. **Referral System Management**
1. Assign referral codes to MRs automatically
2. Define rules:
    a. Retailer onboarding = Fixed commission
    b. First order = Percentage-based incentive
    c. Repeat orders = Variable commission

3. Performance analytics of referral-based sales

ix. Document Management

1. Document repository for all KYC data
2. Digital verification process with approval timestamps
3. Notifications for expired Drug Licenses or missing files

x. Marketing & Communication Tools

1. In-app banner management
2. Create and schedule push notifications
3. Scheme management (e.g., "Buy 10 Get 1 Free")
4. Highlight top-performing products
5. Email/SMS campaign integration

xi. Reports & Analytics

1. Comprehensive real-time analytics:
    a. Sales by region/product/agent
    b. Commission reports
    c. Inventory flow and reorder points
    d. Payment reconciliation
    e. Retailer activity logs
2. Export data as CSV or PDF

xii. System Configuration

1. Manage app-level preferences
2. Tax & GST configuration
3. Payment gateway integration (Razorpay / Paytm / Phonepay etc.)
4. Manage email templates, notification templates
5. Control user access rights and audit logs

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b. Retailer App (B2R Channel)

i. Registration & Verification

1. Sign-up Form:
    a. Business Name, Owner Name, Contact Info
    b. Address and Pincode (auto-mapped via Google API)
    c. Document uploads:
        i. Drug License
        ii. GST Certificate
        iii. PAN Card
    d. Optional MR Referral Code
2. Email & Mobile OTP verification
3. Status Tracking:
    a. "Pending Approval" → "Approved" → "Active"
4. Admin approval required before activation

ii. Onboarding Flow

1. Welcome guide for first-time users
2. Dashboard highlights: Orders, Offers, Wallet
3. Notification for approval/rejection status

iii. **Product Catalog**
    1. Smart search, filters, and sorting options
        a. Category, Price, Brand, Composition
    2. Product details:
        a. Description, Composition, Pack Info, MRP, Schemes
    3. Real-time stock visibility
    4. Download product sheets

iv. **Cart & Order Flow**
    1. Add to cart, wishlist, bulk order option
    2. Apply coupons/offers
    3. View order summary with taxes and shipping details
    4. Payment options:
        a. Online (UPI, Net Banking, Credit/Debit)
        b. Order confirmation with invoice download

v. **Order Management**
    1. Order tracking (status updates via push notifications)
    2. Repeat order feature
    3. View complete purchase history
    4. Raise replacement or refund request (will be initiated after admin approval and only after submitting the required docs or images)

vi. **Offers & Promotions**
    1. Exclusive B2R offers
    2. Product bundle deals
    3. In-app push notification for limited-time discounts

vii. **Wallet & Credit**
    1. Digital wallet for refunds or prepayments
    2. Wallet top-up via UPI
    3. Transaction logs with date and status

viii. **Support Section**
    1. Ticket raising system
    2. Chat support with MR or Lucent Biotech Support
    3. FAQs, help center

ix. **Notifications**
    1. Real-time order and delivery alerts
    2. New product launch updates
    3. License renewal reminders

x. **Profile Management**
    1. Update personal & business details
    2. Upload revised documents
    3. View verification and approval logs

c. *Agent (MR) App*

i. **Registration & Verification**
    1. Fields:
        a. Full Name, Mobile, Email, Territory, Employee ID
        b. Upload ID Proof
        c. Bank Details (for commission transfer)

   d. Referral Code (if any)
2. OTP verification for authentication
3. Admin approval required for activation

ii. Dashboard

1. **Overview:**
   a. Total Retailers Referred
   b. Commission Summary (Earned / Pending)
   c. Monthly Orders by Retailers
   d. Personal Performance Graph
2. **Retailer Onboarding**
   a. Ability to register new retailers via MR app
   b. Fill retailer form and upload documents on their behalf
   c. Track onboarding status (Pending / Approved / Rejected)
3. **Referral System**
   a. Unique referral code and link generation
   b. Share via SMS, WhatsApp, or QR
   c. Track performance and conversion rate
   d. Earn commission on retailer onboarding and order milestones
4. **Commission Tracking**
   a. Real-time view of commissions
   b. Commission logs by order/date
   c. Option to request payout
   d. Notifications on credited commissions
5. **Product Promotion Tools**
   a. Digital catalog and product brochure sharing
   b. Share offers/schemes with retailers
   c. Push notifications for new launches
6. **Performance Targets**
   a. Monthly/quarterly sales goals
   b. Leaderboard comparison with other MRs
   c. Incentive-based gamification
7. **Notifications & Alerts**
   a. New scheme updates
   b. Commission disbursal alerts
   c. Retailer approval updates
8. **Support**
   a. Raise issues or queries
   b. Chat with admin or finance for commission clarifications

7. Marketing Enablement

a. In-App Banners & Campaigns: Showcase Lucent's top-performing brands or new products.
b. Referral Program: Drive organic user acquisition through MR-led retailer onboarding.
c. Push Notifications: Personalized campaigns for reorders, offers, and launches.
d. Data Analytics: Identify potential high-performing regions and product categories.
e. MR Performance Reports: Reward top performers with bonuses or recognition.

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8. Technology Stack

Layer	Technology
Frontend (App)	Flutter (Cross-platform for Android & iOS)
Backend	Node.js
Database	PostgreSQL (AWS RDS)
Hosting	AWS EC2 + S3 + CloudFront
Auth	JWT Tokens
Payments	Razorpay / Paytm / PhonePay etc.
Notifications	Firebase Cloud Messaging + AWS SNS
Analytics	Google Firebase Analytics / AWS QuickSight
Security	SSL, AES Encryption, AWS WAF

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9. Approval Workflows

Process	Trigger	Approval Role	Outcome
Retailer Registration	Document submission	Admin	Access to app
MR Registration	Bank + ID submission	Admin	Portal access
Commission Payout	Order completion	Admin/Finance	Credited to MR
Product Listing	Product entry	Admin	Visible to retailers
Offer Activation	Marketing Campaign	Admin	Shown in app

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10. AWS Infrastructure & Maintenance

Bhoomi Techzone Pvt. Ltd. will manage:

a. EC2, RDS, S3, CloudFront setup and configuration
b. SSL implementation and firewall setup
c. CI/CD pipeline for deployment
d. Data backups and restoration
e. 24x7 uptime monitoring
f. Performance and scaling optimization

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11. Commercials & Project Timeline

a. **Project Cost**
    i. The total cost for the development, deployment, and delivery of the B2R Medical Application is
        1. **Total Project Cost: INR 3,00,000 + 18% GST (Three Lakh Only + Applicable GST)**
        2. **Total with GST INR 3,54,000**
b. **Payment Structure (Milestones)**
    i. The payment will be divided into four structured milestones, aligned with clear deliverables:
        1. **Milestone 1 – 20% (INR 70,800)**
            a. Due at Project Initiation
                i. Project kick-off
                ii. Requirements finalization

iii. Technical architecture + UI/UX planning
iv. Project setup on AWS/Git

2. **Milestone 2 – 30% (INR 1,06,200)**
   a. Due upon Completion of Core Modules
      i. Retailer App (Registration, Login, Catalog, Cart, Basic Orders)
      ii. Agent/MR App (Registration, Referral Flow)
      iii. Admin Portal: User & Product Management
      iv. API Integrations & Database Setup
3. **Milestone 3 – 30% (INR 1,06,200)**
   a. Due upon Completion of Advanced Modules
      i. Order Management (Tracking, Invoices, Return/Refund Flow)
      ii. Commissions Module (MR Payout, Tracking)
      iii. Offers, Schemes & Marketing Tools
      iv. Notifications, Wallet, Support Tickets
      v. Full AWS Deployment (EC2, RDS, S3, CloudFront)
4. **Milestone 4 – 20% (INR 70,800)**
   a. Due at Final Delivery
   b. UAT (User Acceptance Testing)
   c. Bug fixes & polishing
   d. Documentation handover (Admin + MR + Retailer)
   e. Live deployment on production
   f. 45 days of post-launch support

c. **Timeline: 1.5 - 2 Months**
   i. Breakdown:
      1. Weeks 1–2: Architecture, UI/UX, setup
      2. Weeks 3–6: Core development for all 3 roles
      3. Week 7: Payment, notifications, commission engine, marketing tools
      4. Week 8: UAT, bug fixes, optimization, final deployment

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12. Future Enhancement Scope (Phase 2)

a. ERP and CRM integration for stock synchronization
b. AI-driven reorder suggestions
c. Delivery route optimization
d. Multi-language interface
e. Automated invoicing & GST filing
f. MR geo-tracking for territory management
g. Retailer loyalty program